Assignment 1: QA System and Knowledge Graph Specification

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1. What are the essential skills required for a data engineer?

Classification: Non-factual

Justification: The word "essential" is subjective. Without a strict mathematical rule or threshold (like "skills mentioned in 80% of job postings"), deciding what is essential requires human judgment and evaluation, not just simple data retrieval.

Factual Sub-questions:

1. Which skills appear most frequently in the company's job postings for data engineers?
2. Which skills are officially linked to the data engineer role in the ESCO framework?

2. Which employees have at least five years of experience in project management?

Classification: Factual

Justification: This asks for existing, historical facts. We can answer it by simply filtering the HR database for employees whose objective work history in project management is greater than or equal to five years.

3. How has demand for cybersecurity professionals in Europe changed over the past five years?

Classification: Factual

Justification: We can measure "demand" objectively by simply counting the historical number of open job positions for cybersecurity professionals per year over the last five years and calculating the numerical difference.

4. Which occupations are likely to experience the greatest growth in demand over the next decade?

Classification: Non-factual

Justification: This question asks for a prediction about the future ("likely to experience", "next decade"). It cannot be answered by just retrieving existing facts; it requires predictive modeling and assumptions about market trends.

Factual Sub-questions:

1. Which occupations had the highest growth rate in job openings over the last 3 years?
2. Which occupations are projected to grow according to the latest official EU labor market report?

5. If Maria completed a certification in cloud computing, which additional job openings would she qualify for?

Classification: Factual

Justification: This is a hypothetical scenario, but it is entirely deterministic. If we temporarily add the certification to Maria's profile in the database, we can use a strict rule-based match to see exactly which new job requirements she now meets.

6. Which university degrees are most commonly held by software engineers in Germany?

Classification: Factual

Justification: This requires a simple statistical calculation (finding the mode or highest frequency) of the degrees currently held by software engineers in Germany, based entirely on existing historical data.

7. How would the company's productivity change if all employees worked four days per week?

Classification: Non-factual

Justification: This is a complex hypothetical scenario. "Productivity" can be hard to define, and predicting how humans will behave under new conditions requires causal inference and subjective assumptions, not just data retrieval.

Factual Sub-questions:

1. What was the average measurable output per employee last year?
2. Did the company run any 4-day workweek pilot programs before, and what was their measured output?

8. Which employees are most likely to leave the company within the next year?

Classification: Non-factual

Justification: This asks for a future prediction. It requires running probabilistic machine learning models to guess human behavior, which means the answer is an estimation, not a guaranteed existing fact.

Factual Sub-questions:

1. Which employees have been in the exact same role for over 4 years without a promotion?
2. What is the historical attrition rate for each department in the company?

9. How does the average salary of data scientists compare with that of data analysts across European countries?

Classification: Factual

Justification: This is based on a straightforward mathematical calculation. We can calculate the average salaries from existing numerical data for both roles and simply compare the two numbers.

10. If AI automated half of the routine tasks performed by accountants, which tasks would remain part of their occupation?

Classification: Non-factual

Justification: The definition of a "routine task" is subjective. Deciding which exact tasks an AI can automate requires qualitative judgment and assumptions about technology capabilities, not just querying a database.

Factual Sub-questions:

1. What are all the distinct tasks currently linked to the accountant role in the knowledge graph?
2. Which accounting tasks are officially classified by the company as "strategic" rather than "routine"?

11. Which training programs cover the skills required for a marketing specialist to transition into data analytics?

Classification: Factual

Justification: This can be answered using standard set operations. We find the skills required for data analytics, subtract the skills the marketing specialist already has, and then search the database for training programs that teach those exact missing skills.

12. How many employees moved from technical to managerial positions in the past three years?

Classification: Factual

Justification: Assuming roles are clearly tagged as "technical" or "managerial" in the system, this requires a simple aggregation and counting of historical role-change events in the HR database.

13. Which employees are ready for promotion to a senior management position?

Classification: Non-factual

Justification: Being "ready for promotion" involves a subjective evaluation of leadership skills, soft skills, and managerial reviews. Unless the company has strict, purely mathematical criteria for promotions, this requires human judgment.

Factual Sub-questions:

1. Which employees meet the minimum required years of experience for a senior management role?
2. Which employees received the highest performance scores in the last two review cycles?

14. Which occupations offer the best career prospects for someone with a degree in mathematics?

Classification: Non-factual

Justification: "Best career prospects" is highly subjective. It could mean the highest salary, the best work-life balance, or the lowest unemployment rate. It cannot be answered without explicitly defining what "best" means.

Factual Sub-questions:

1. Which occupations requiring a mathematics degree have the highest average salary?
2. Which of these occupations have the highest number of open job postings?

15. Has the introduction of AI tools caused an increase in the productivity of software developers in the company?

Classification: Non-factual

Justification: The word "caused" implies a causal relationship. Proving that AI tools (and not something else) increased productivity requires complex statistical analysis and assumptions to isolate variables. It is not a simple fact lookup.

Factual Sub-questions:

1. How many pull requests were merged on average by developers before and after adopting AI tools?
2. What is the current adoption rate of AI tools among the company's developers?

16. Which employees are at risk of becoming redundant due to automation?

Classification: Non-factual

Justification: Identifying who is "at risk" requires predictive modeling and assumptions about how fast technology will evolve and how it will affect specific human tasks in the future.

Factual Sub-questions:

1. Which employees spend more than 70% of their working hours on manual data entry?
2. Which roles contain tasks that have already been mapped to existing automation software in the company?

17. Would hiring more junior developers instead of senior developers reduce the company's software development costs?

Classification: Non-factual

Justification: While salary differences are factual, total "development costs" include onboarding time, code quality, and time spent fixing bugs. Calculating this requires complex hypothetical assumptions, not just adding up salaries.

Factual Sub-questions:

1. What is the exact difference in average salary between a junior and a senior developer?
2. How many hours on average do senior developers spend reviewing code written by juniors?

18. How has the number of applications for different occupations changed over the past five years?

Classification: Factual

Justification: This is a straightforward retrieval of historical data. We just count the number of job applications per occupation for each year over the last five years and calculate the trend.

19. Which training programs have been most effective in helping employees transition into new roles?

Classification: Non-factual

Justification: "Most effective" is subjective and requires defining success metrics. Also, proving that the training program directly caused the transition involves complex analysis, not just querying facts.

Factual Sub-questions:

1. What percentage of employees who completed Program X changed roles within 6 months?
2. What is the successful completion rate for each training program offered by the company?

20. Which skills should a data analyst prioritize learning to remain employable over the next decade?

Classification: Non-factual

Justification: This question asks for advice and predictions about the distant future ("should prioritize", "next decade"). It relies on guessing market trends and cannot be determined purely from current facts.

Factual Sub-questions:

1. Which emerging skills had the highest percentage growth in data analyst job postings over the last two years?
2. Which technical skills co-occur most frequently with "Data Analyst" in current tech industry reports?

AI Use Declaration

I used an AI to help me with this assignment. First, I asked the AI to explain the exact difference between 'factual' and 'non-factual'. This helped me understand the terms so I could use my own judgment. After that, I asked for the AI's opinion on each of the 20 questions to help me write the explanations and the sub-questions.